

# 1 Locomotion Is a Control Problem

## THE PROBLEM

Position-based locomotion is accurate under known conditions but rigid — it struggles with compliance, disturbance handling, unknown terrain, and the sim-to-real gap.

→ Torque-based control enables compliant, adaptive interaction.

## POSITION CONTROL

Command position



Rigid behavior



Poor adaptation

## TORQUE CONTROL

Command force



Compliant interaction



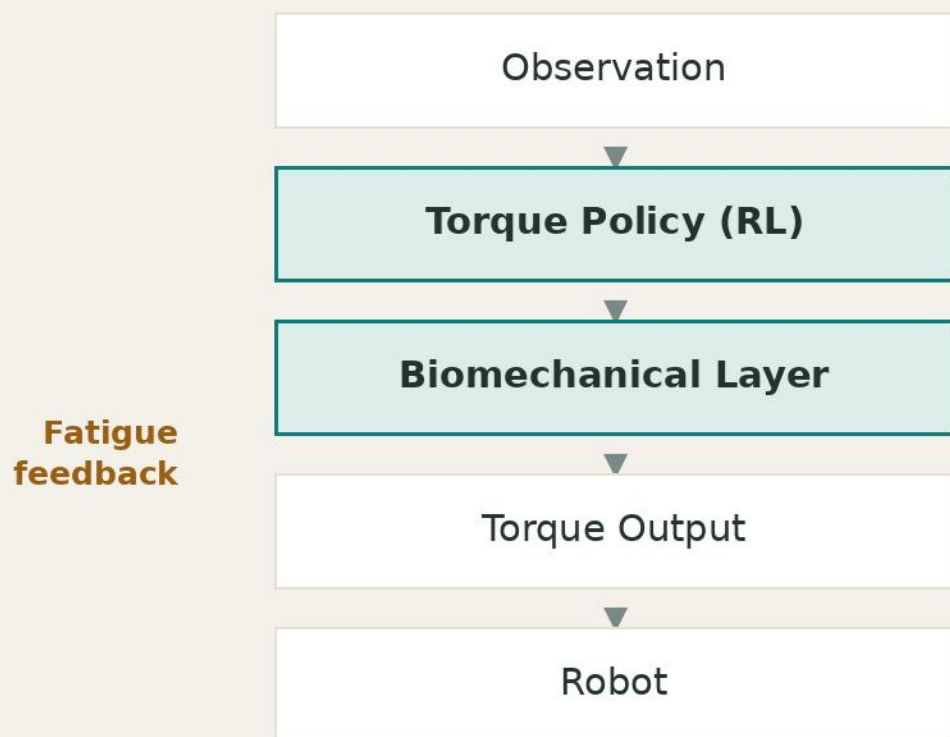
Adaptive response

## RESEARCH QUESTION

**How can robots achieve safer and more adaptive locomotion in unknown environments?**

## 2 SATA: Bio-inspired Adaptation in Torque Control

An RL torque policy wrapped by a biomechanical layer and a growth curriculum.



### Biomechanical Model

activation · muscle · fatigue



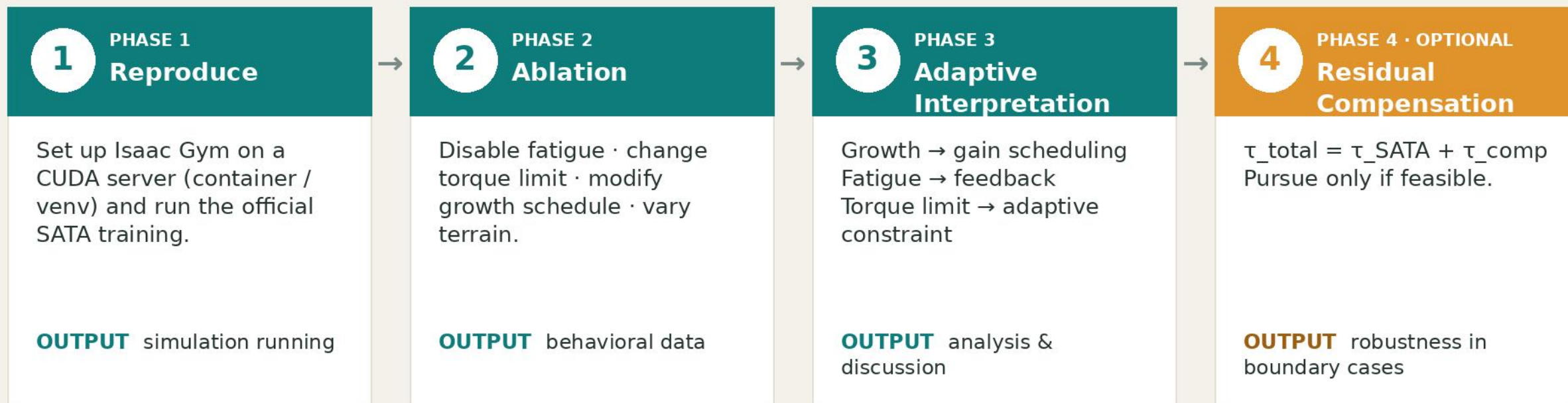
### Growth Mechanism

torque limit · reward · frequency

Goal: smooth, safe torques that progressively unlock capability — deployed zero-shot (sim-to-real).

# 3 Plan: Understand → Reproduce → Analyze

A feasible 3-phase study — we interpret SATA, we do not redesign the RL.



## APPROACH

We understand and analyze adaptive behavior — we do not redesign the RL algorithm.

## 4 Expected Contribution

We study adaptive behavior — not propose a new RL algorithm.

### EXPECTED OUTPUTS

- 1 Simulation reproduction
- 2 Behavior analysis (ablation)
- 3 Adaptive-control interpretation

### RESEARCH QUESTIONS

#### RQ1

Why torque control?

#### RQ2

What creates adaptation?

#### RQ3

How does it relate to adaptive control?

#### RQ4

Can lightweight compensation help?

### IN ONE LINE

This project investigates how bio-inspired torque control creates adaptive locomotion behaviors, and interprets these mechanisms through an adaptive control perspective.